Stochasticity

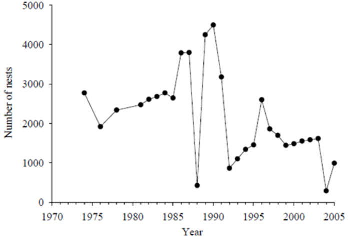

Figure 1. The population trend of Pelagic cormorants at Middleton Island, Alaska from 1974 to 2005 (Hatch et al., unpublished data).

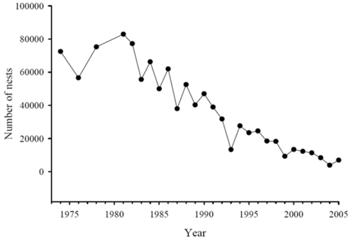

Figure 7. The population trend of black-legged kittiwakes on Middleton Island, Alaska from 1974 to 2005 (Hatch et al., unpublished data).

Learning objectives

1. Introduction
2. Environmental stochasticity
3. Demographic stochasticity

A random variable is a variable whose value can't be predicted with certainty.

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Examples

- Weather
- Our own behavior
- Population size

A random variable is a variable whose value can't be predicted with certainty.

Examples

- Weather
- Our own behavior
- Population size

Is the universe random? Or is just too complex to be predicted with certainty?

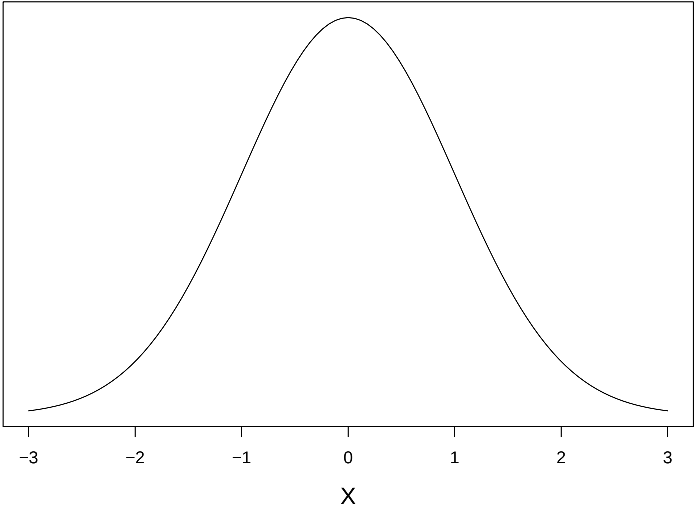
A random variable (X) can be described by a probability distribution.

There are many types of probability distributions

- Normal (or Gaussian)
- Poisson
- Binomial
- Multinomial
- etc. . .

Normal (Gaussian) Distribution

$$X \sim \mathsf{Normal}(\mu = 0, \sigma^2 = 1)$$



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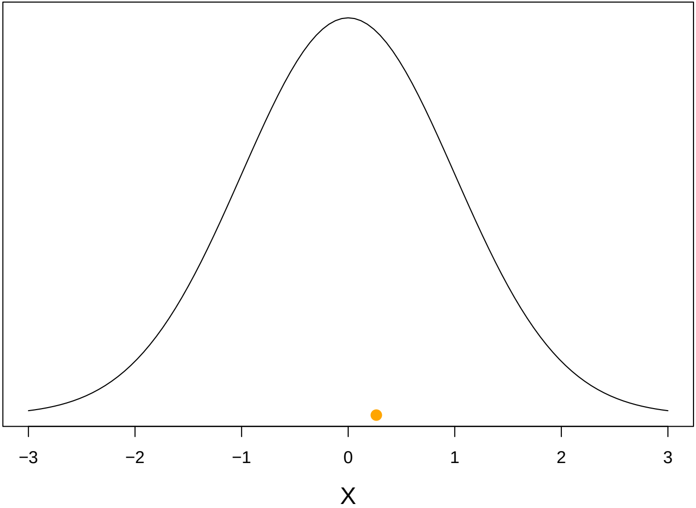

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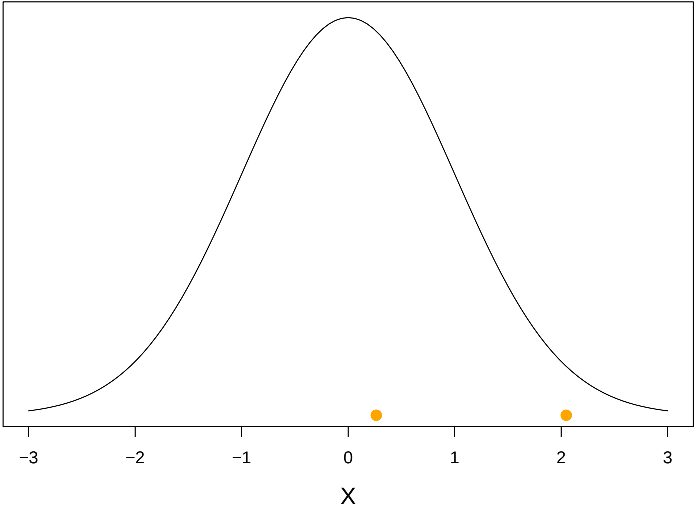

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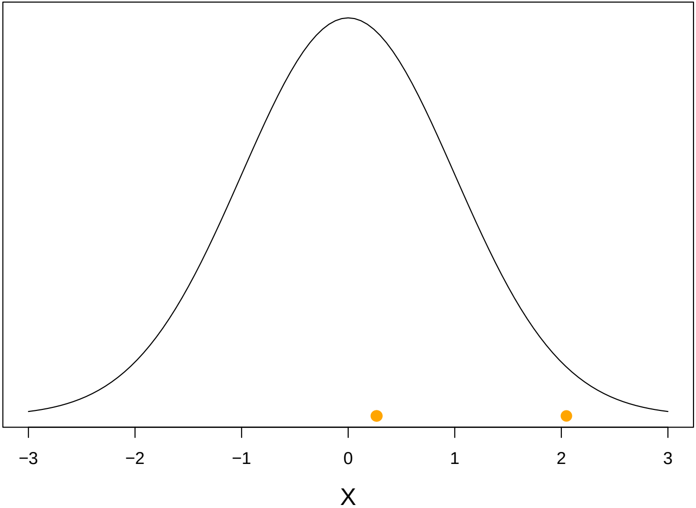

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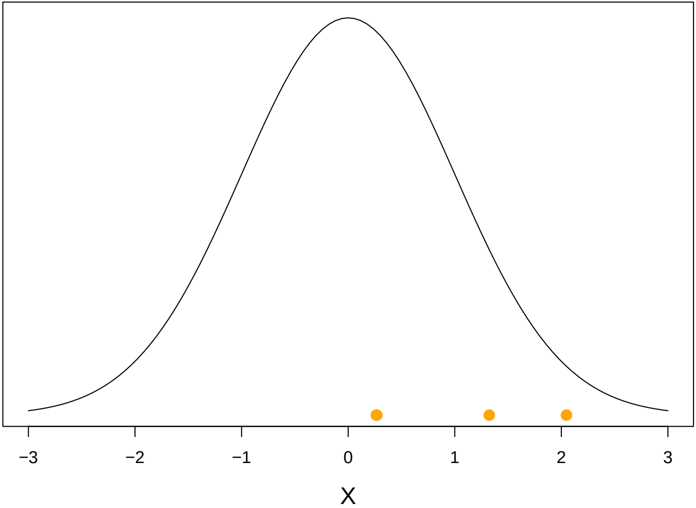

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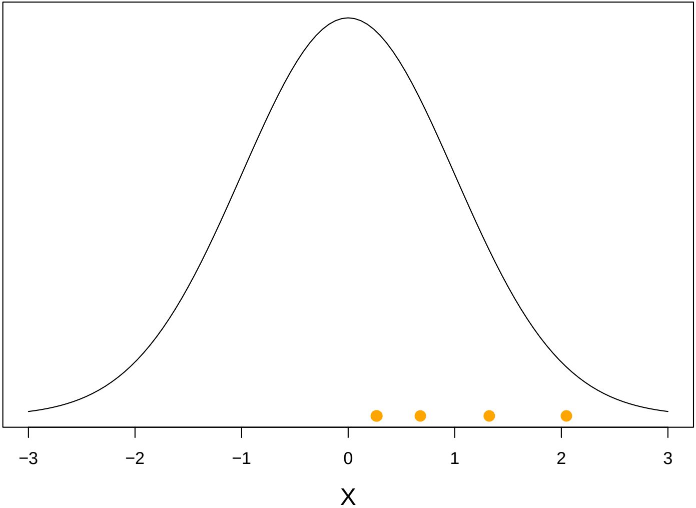

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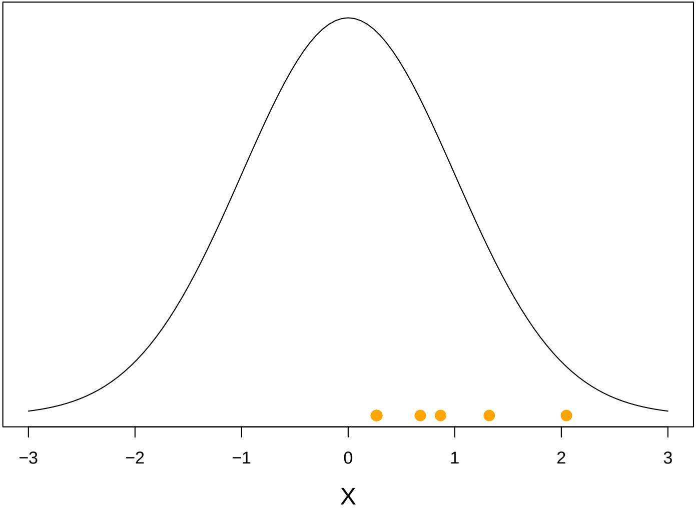

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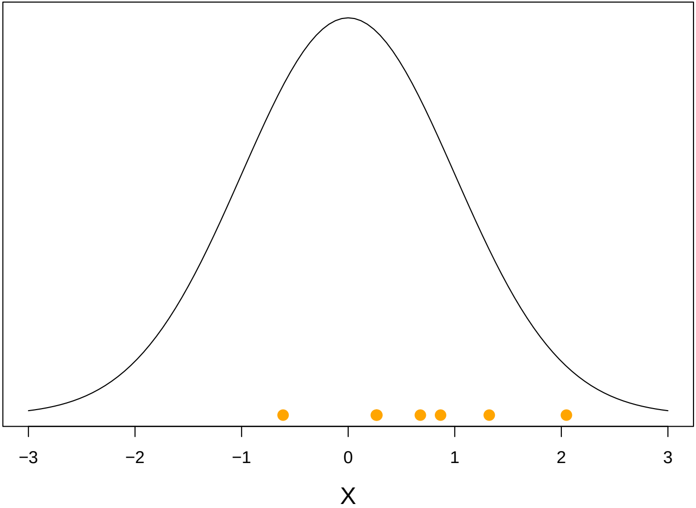

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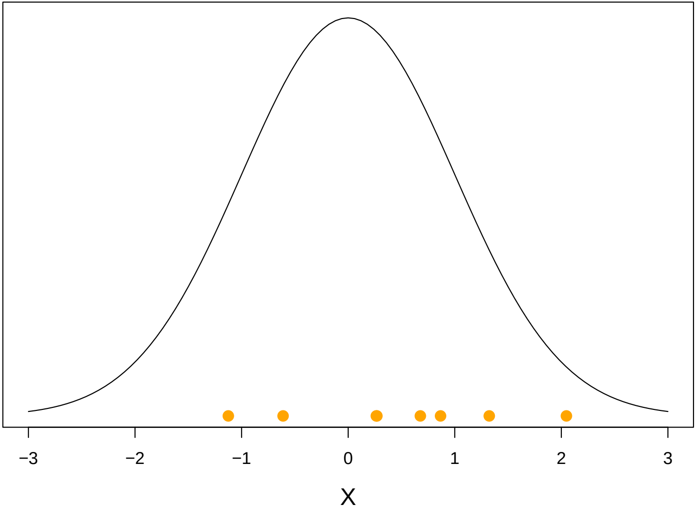

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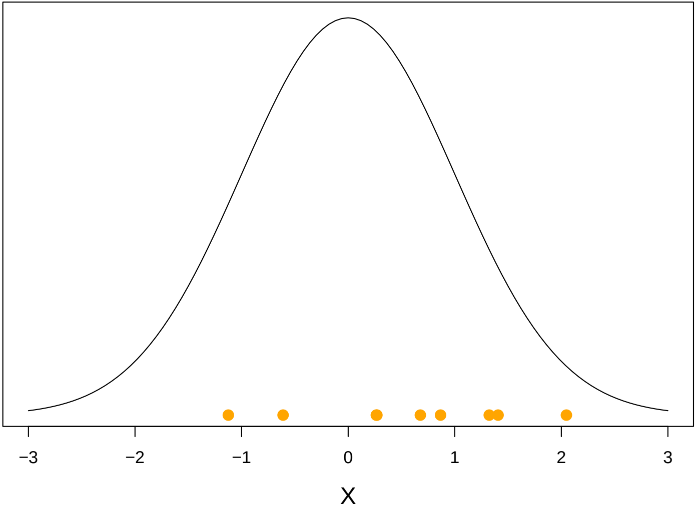

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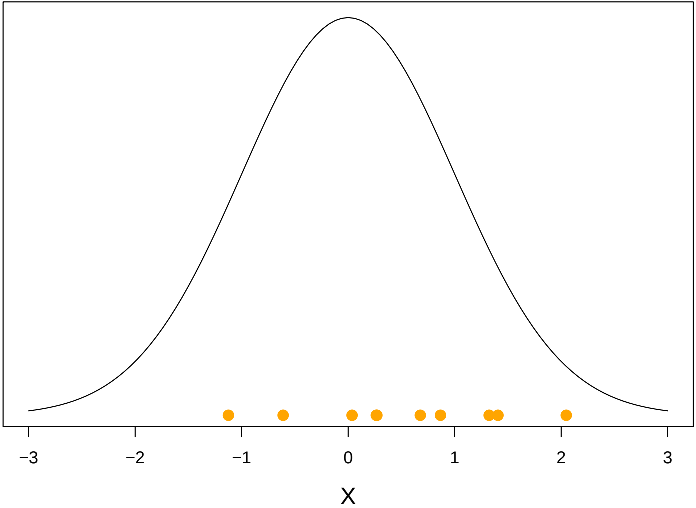
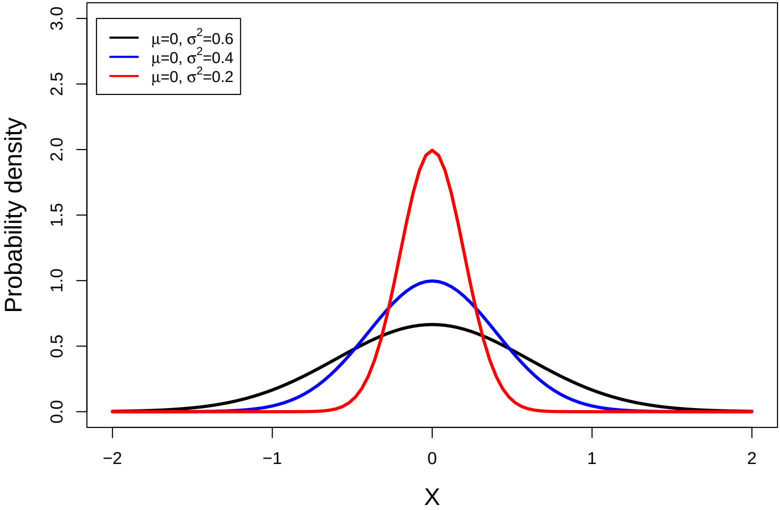
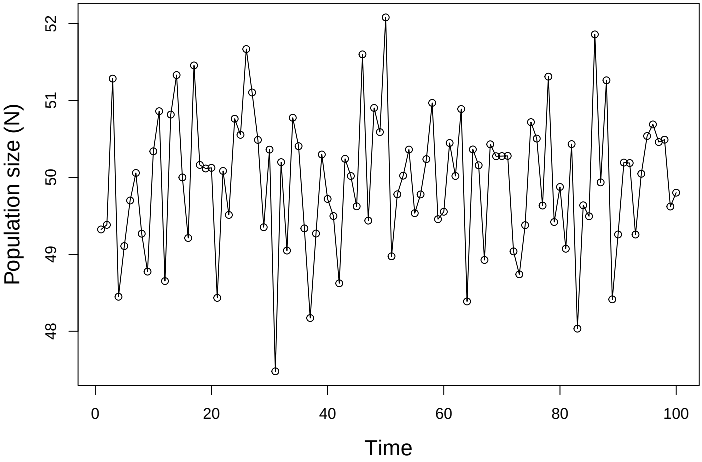



μ=0, σ²=0.6
μ=0, σ²=0.4
μ=0, σ²=0.2
Probability density
0.0
0.5
1.0
1.5
2.0
2.5
3.0
−2
−1
0
1
2
X

$$N_t \sim \text{Normal}(\mu = 50, \sigma^2 = 1)$$



Environmental stochasticity

- Random variation in demographic rates (such as birth and death rates)
- Caused by unpredictable changes (e.g., in weather, habitat, etc. . . conditions) among years

Two important types of stochasticity

Environmental stochasticity

- Random variation in demographic rates (such as birth and death rates)
- Caused by unpredictable changes (e.g., in weather, habitat, etc. . . conditions) among years

Demographic stochasticity

- Random variation in demographic outcomes (such as the number of births and deaths)
- Occurs even when demographic rates are constant

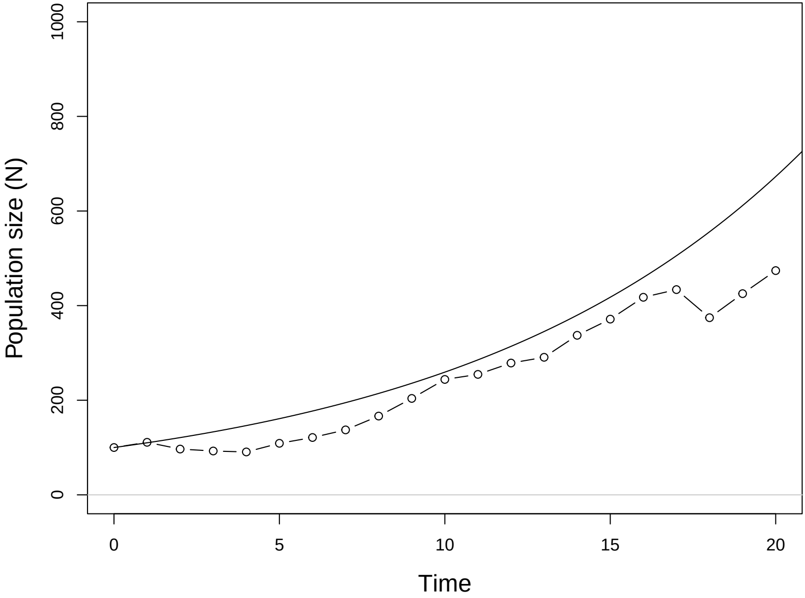
$$N_{t+1} = N_t + N_t r_t$$

$$r_t \sim \mathsf{Normal}(\bar{r}, \sigma^2)$$

R code

```
nYears <- 20
N <- r <- rep(NA, nYears)  ## Create empty N and r
N[1] <- 100                ## Initial value of N
r.bar <- 0.5               ## Average growth rate
sigma <- 0.1               ## StdDev of growth rate
for(t in 2:nYears) {
    r[t-1] <- rnorm(n=1, mean=r.bar, sd=sigma)
    N[t] <- N[t-1] + N[t-1]*r[t-1]
}
```

Example $N_0 = 100,\ \bar{r} = 0.5,\ \sigma^2 = 0.01$



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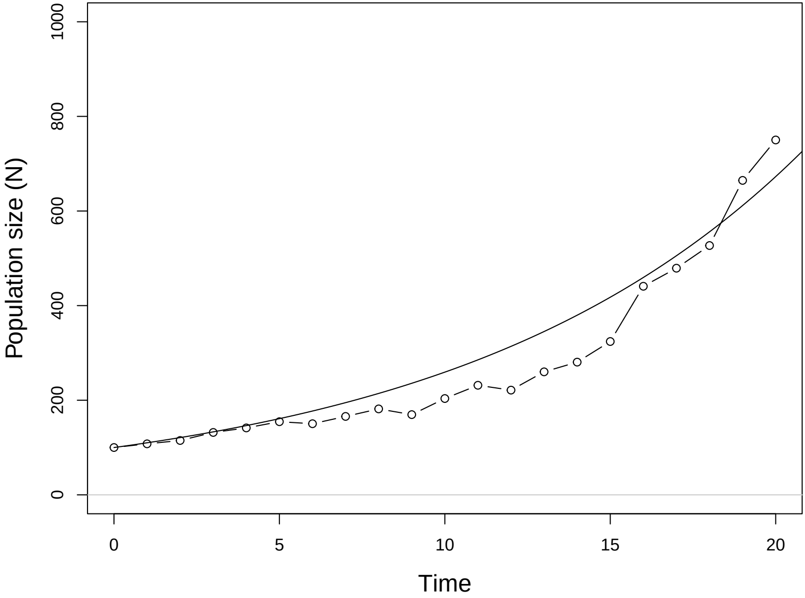

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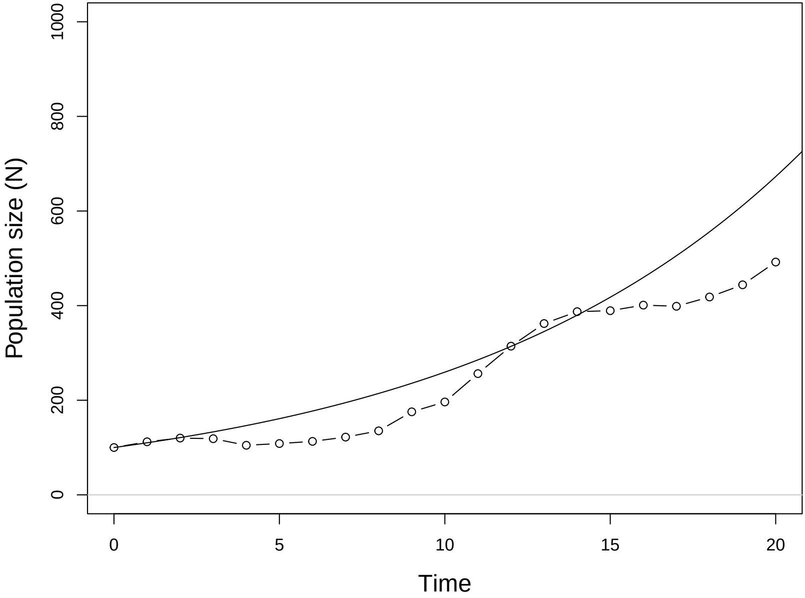

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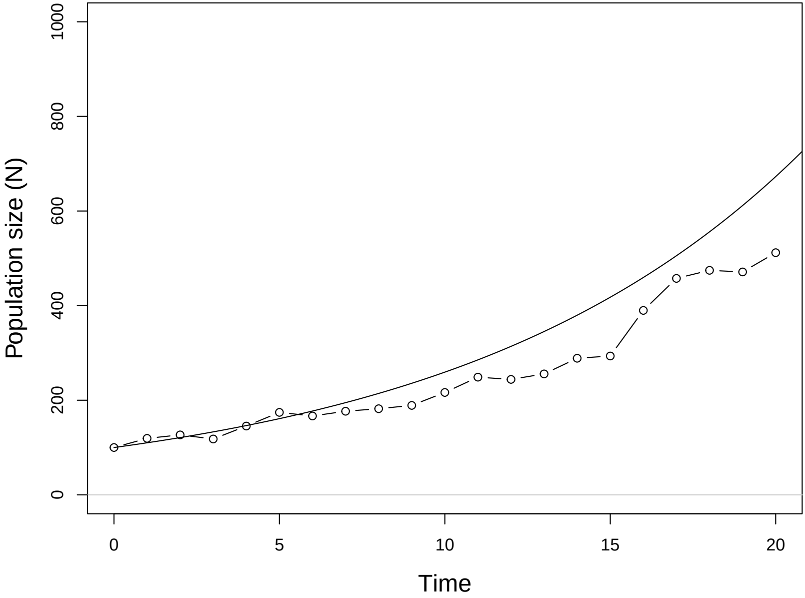

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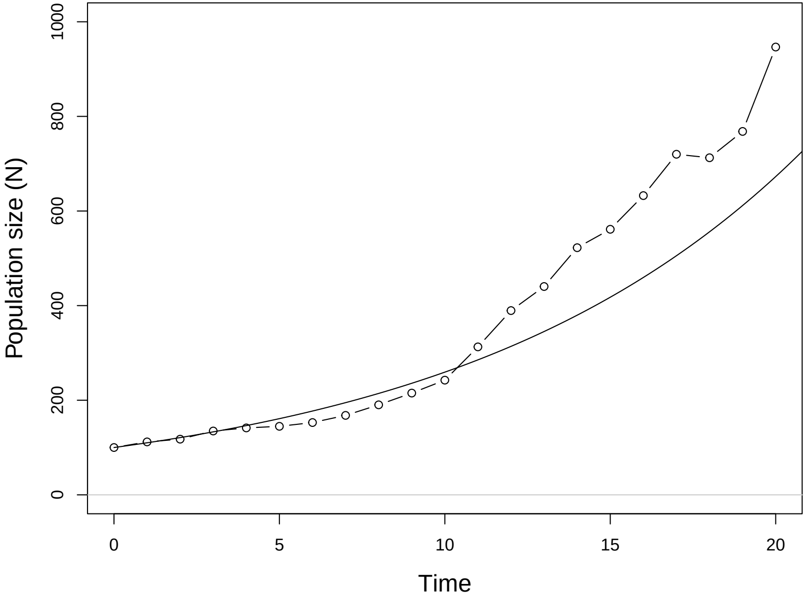

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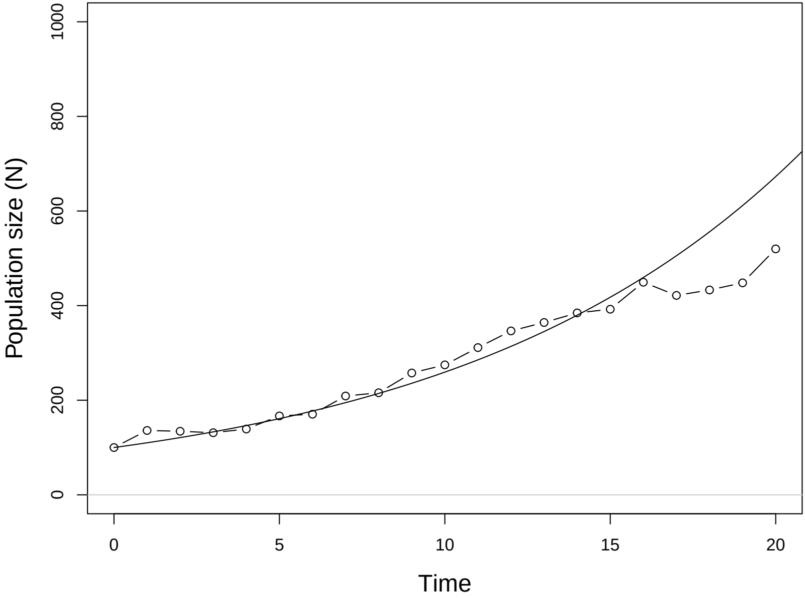

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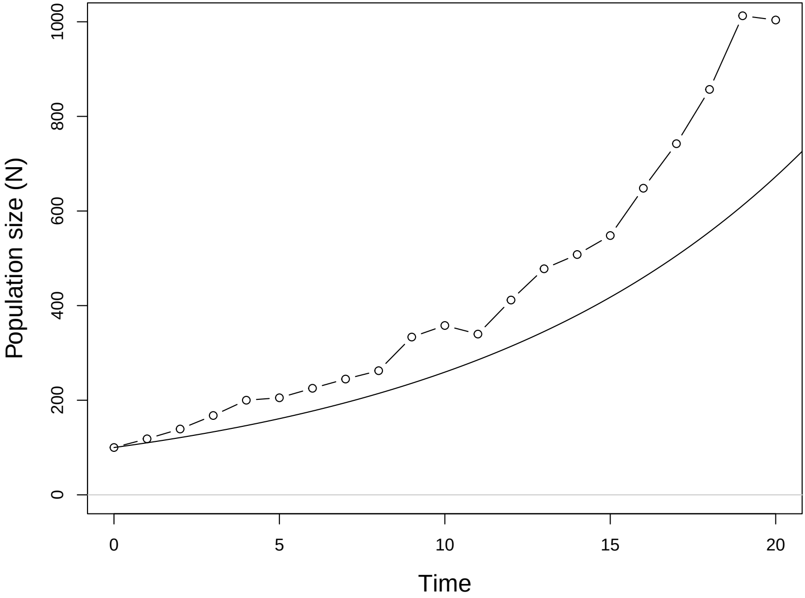

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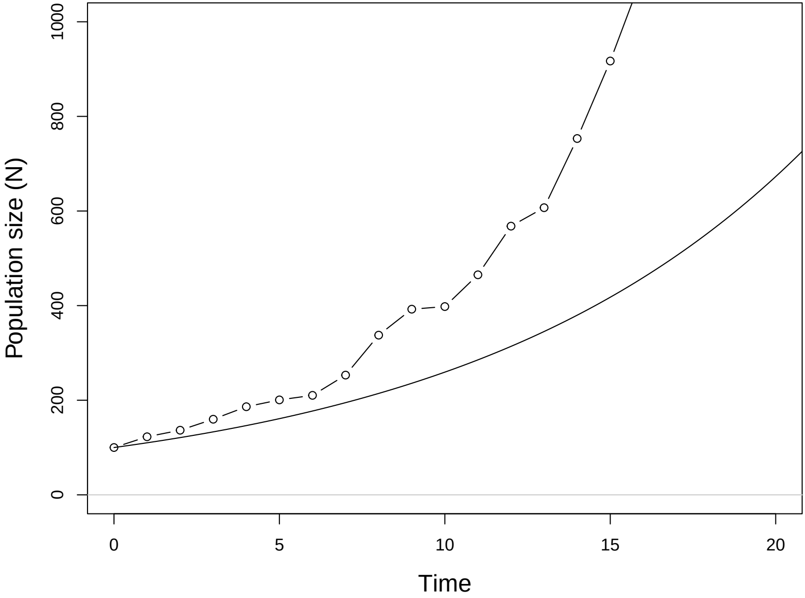

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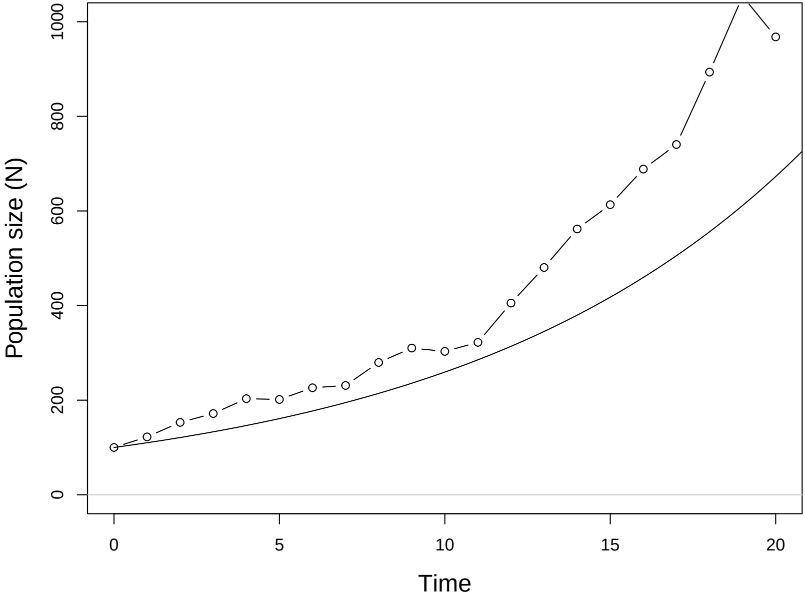

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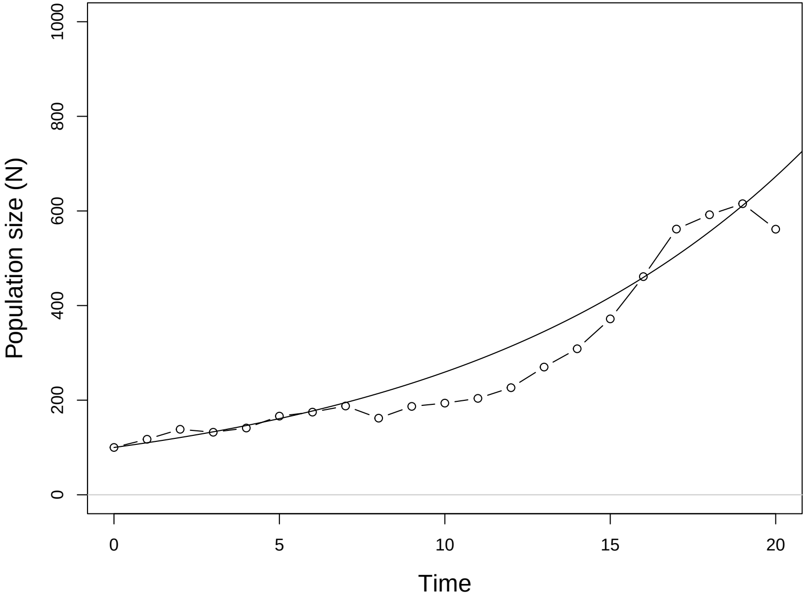

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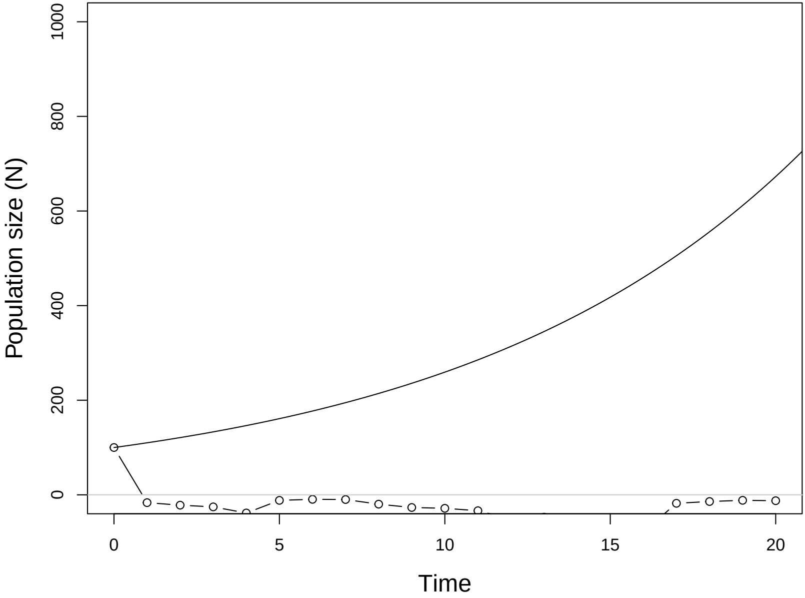

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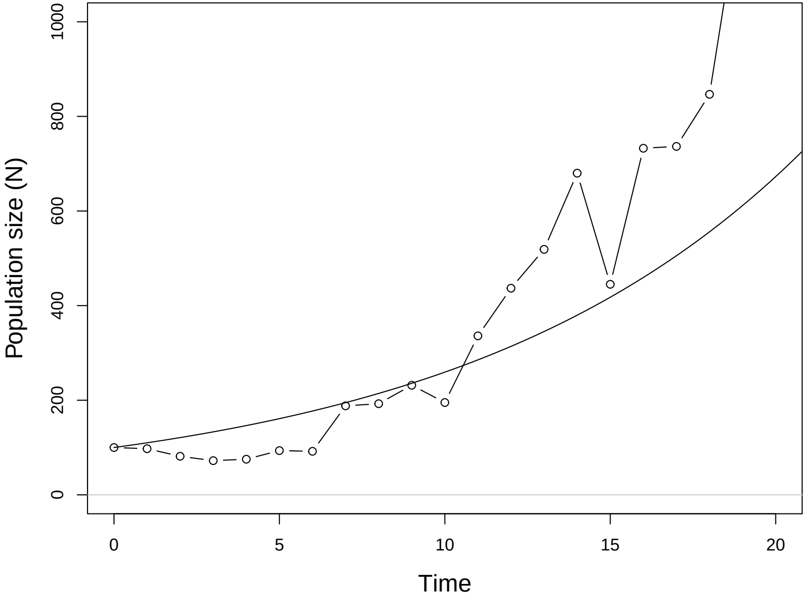

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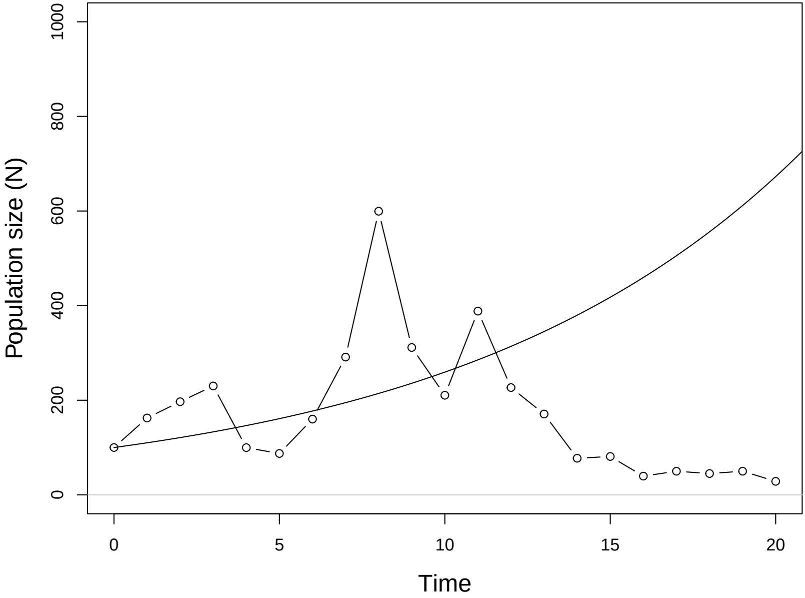

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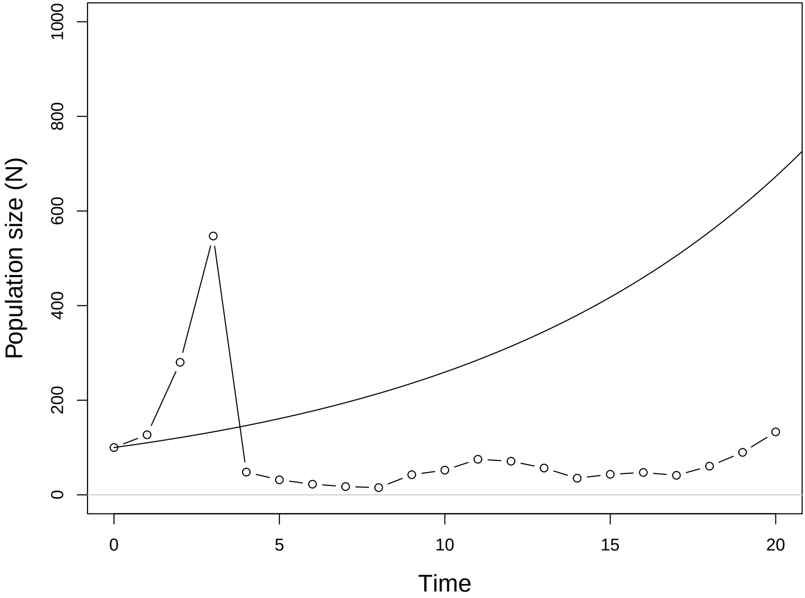

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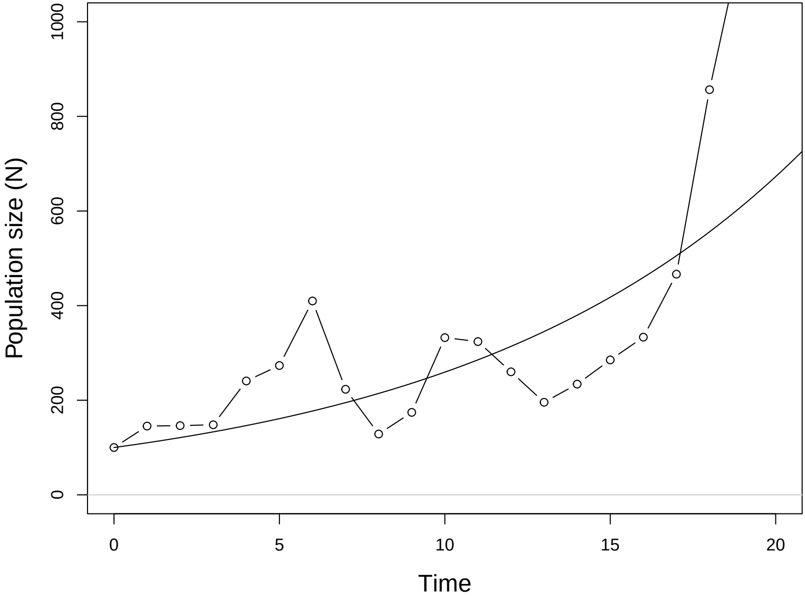

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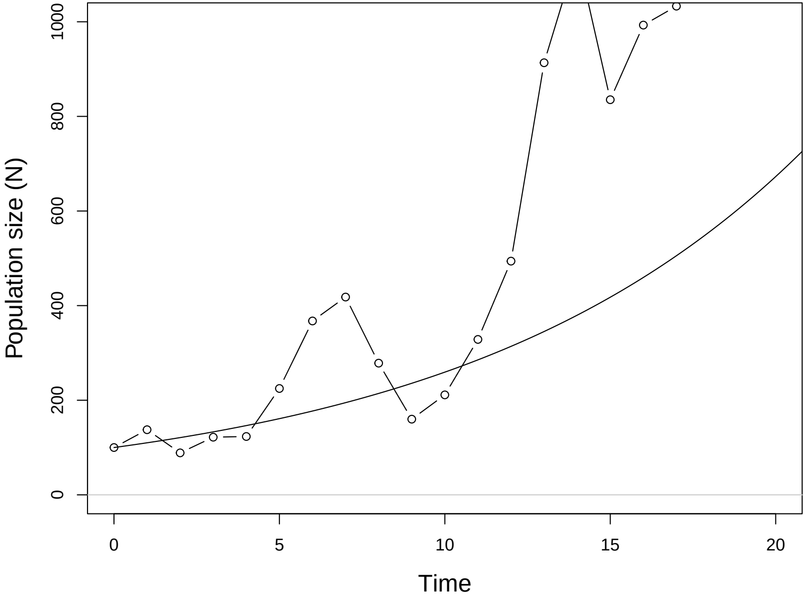

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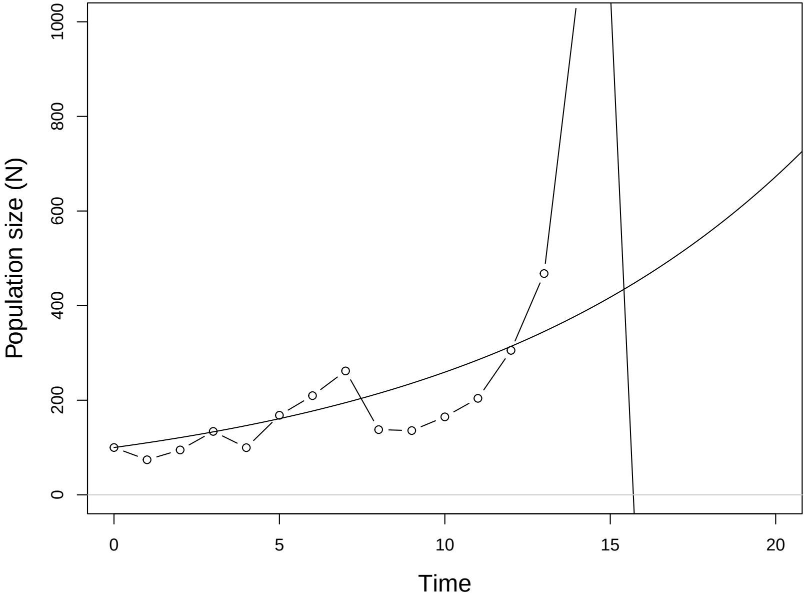

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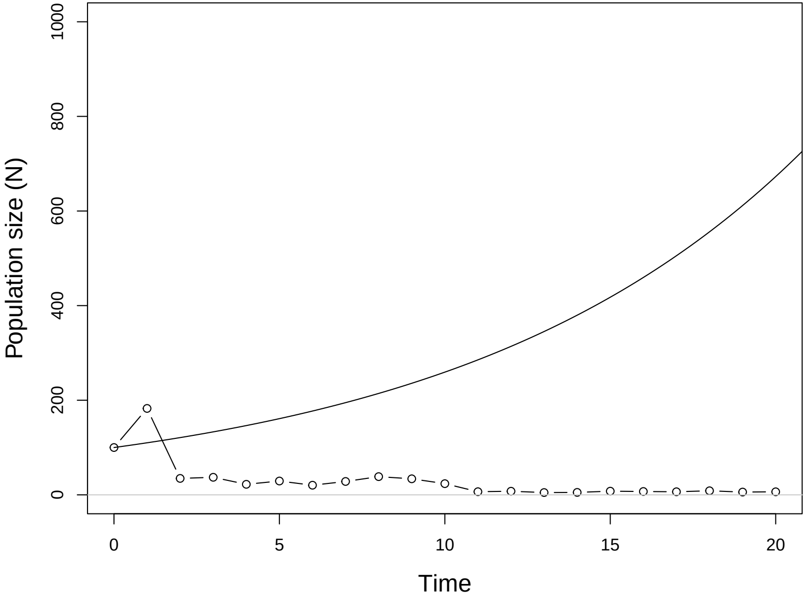

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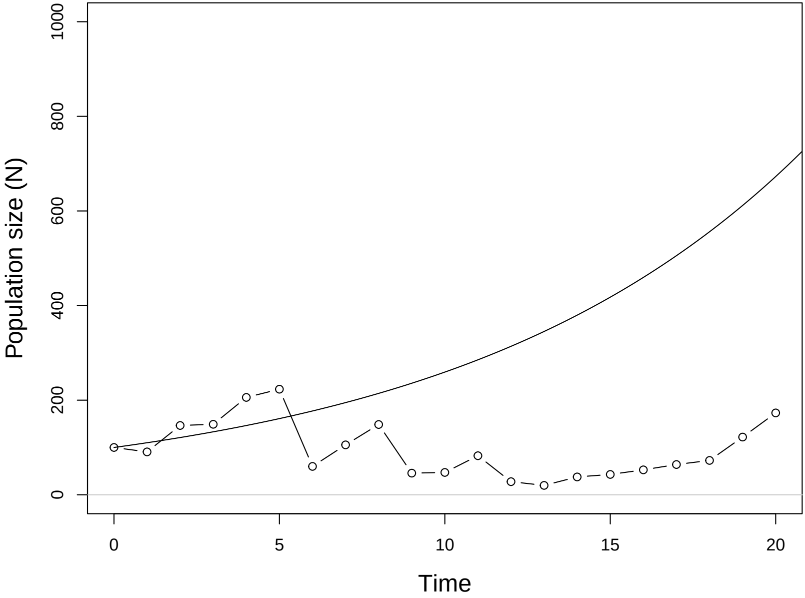

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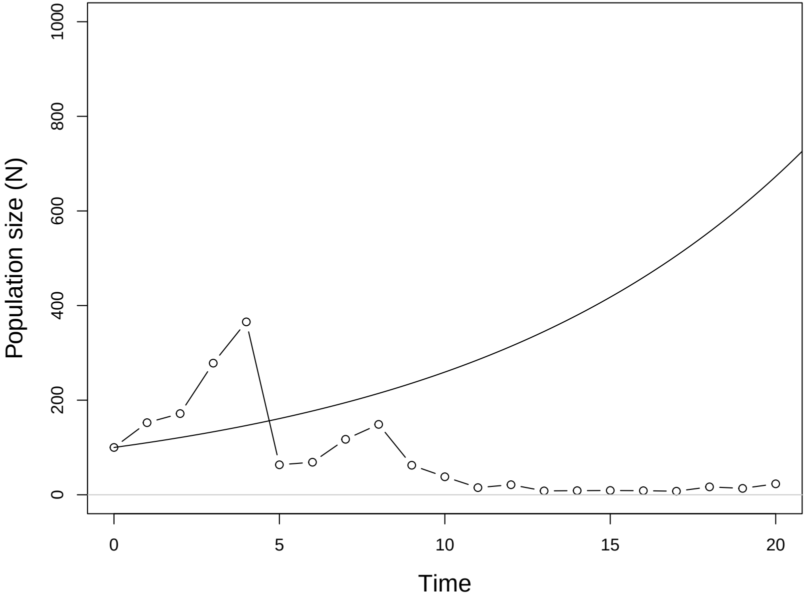
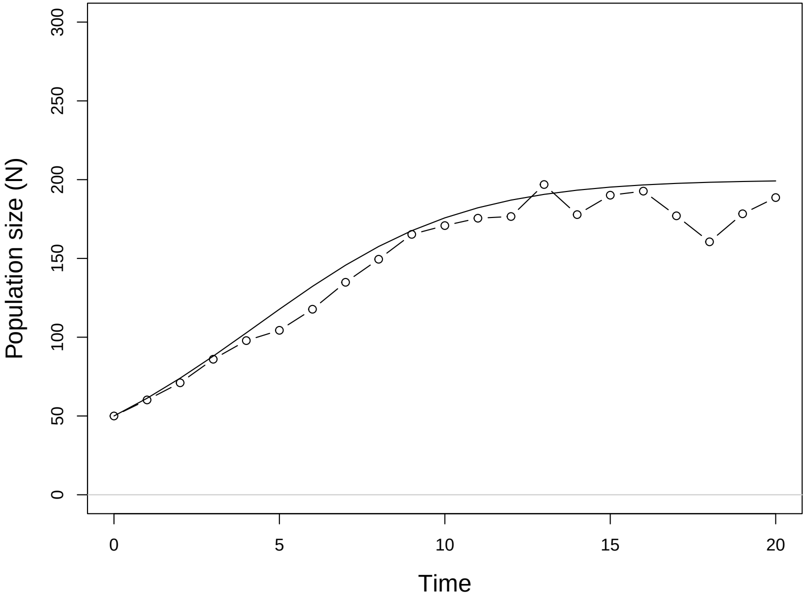


Imagine the carrying capacity fluctuates randomly from year to year.

$$N_{t+1} = N_t + N_t r_{max}(1 - N_t/K_t)$$

$$K_t \sim \mathsf{Normal}(\bar{K}, \sigma^2)$$

Logistic example, $r_{max} = 0.2,\ \bar{K} = 200,\ \sigma^2 = 50$



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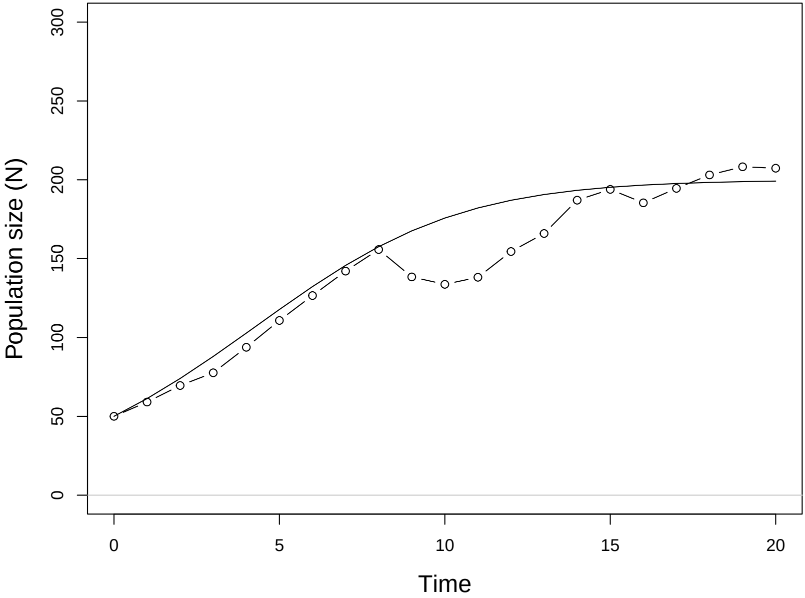

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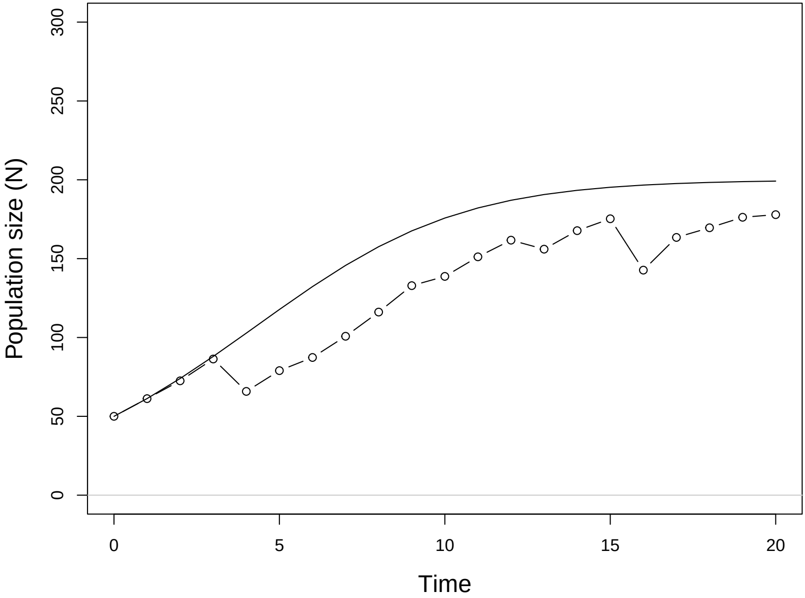

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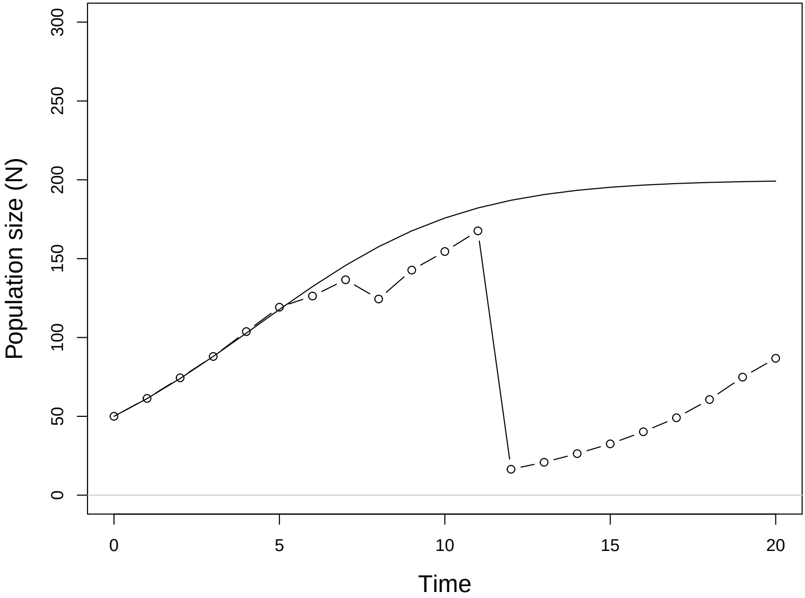

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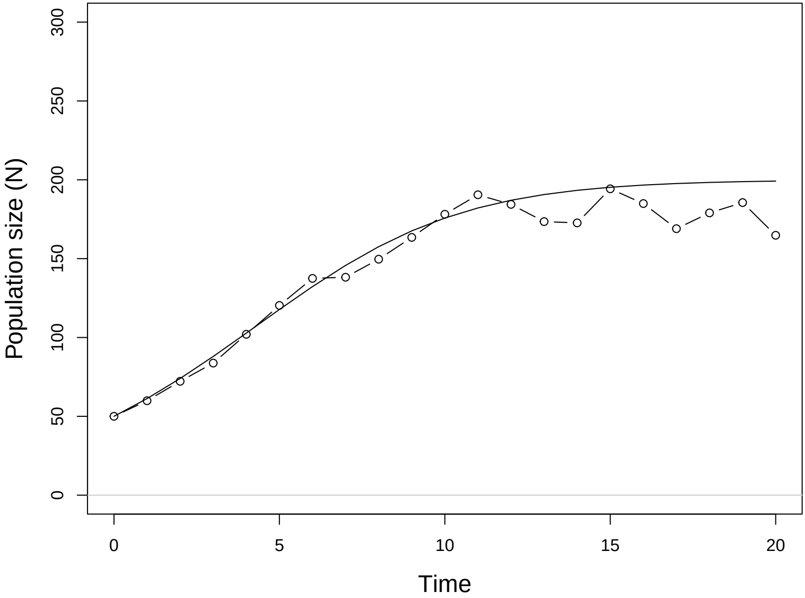

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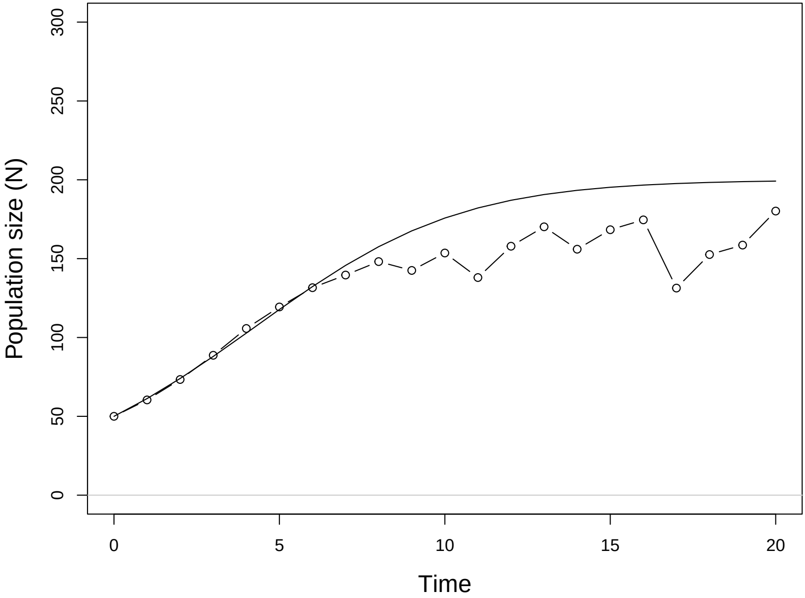

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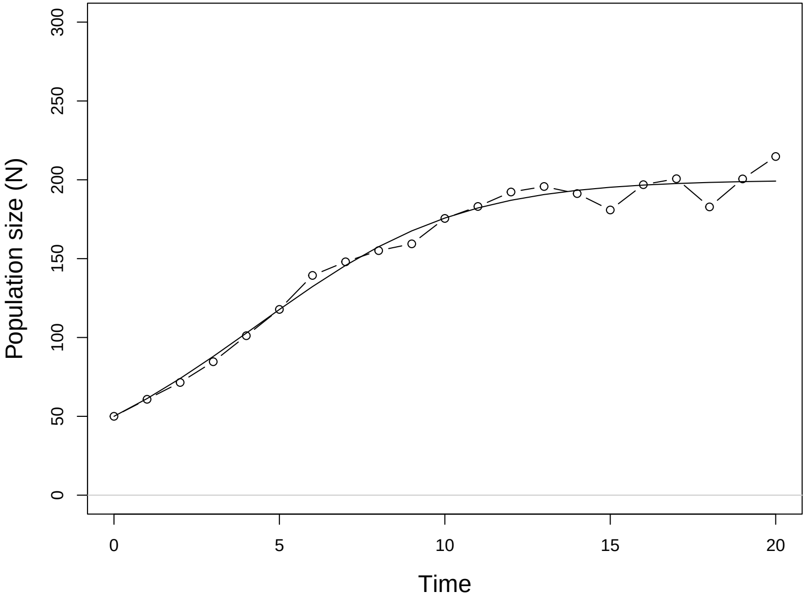

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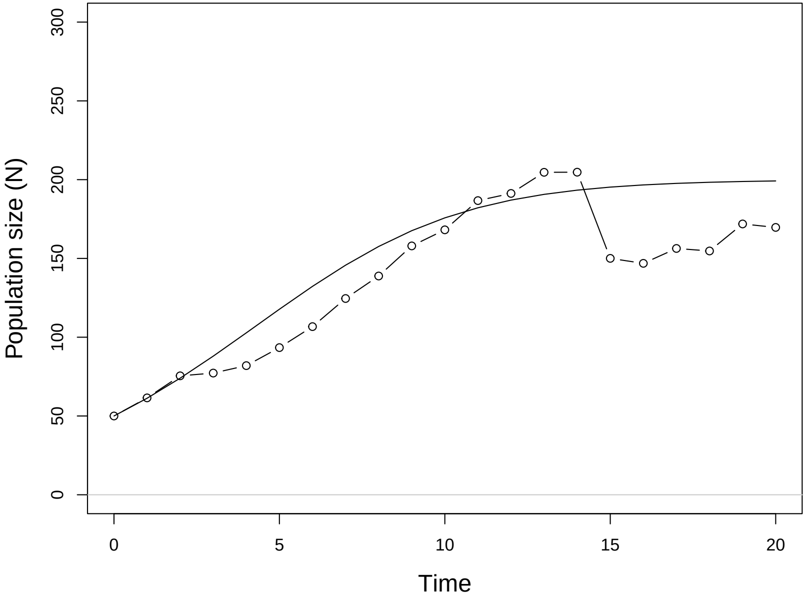

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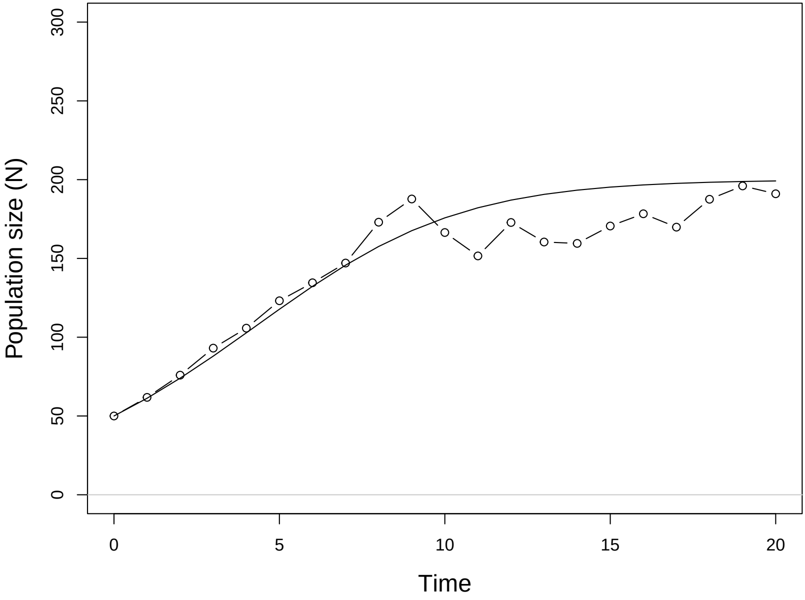

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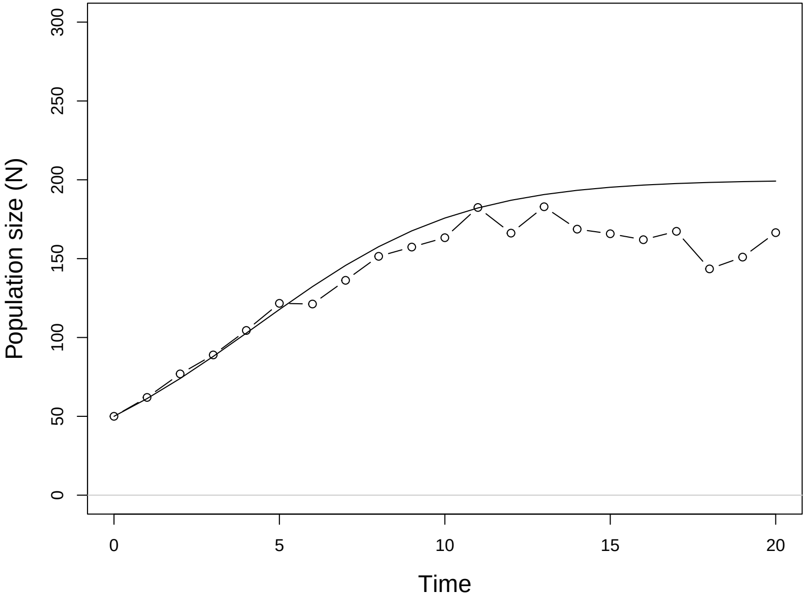

We can make the BIDE model stochastic by modeling random variation in the number of individuals that are born and die each year (we'll ignore movement):

$$N_{t+1} = N_t + B_t - D_t$$

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$$N_{t+1} = N_t + B_t - D_t$$

What distributions should we use for B_t and D_t?

We can make the BIDE model stochastic by modeling random variation in the number of individuals that are born and die each year (we'll ignore movement):

$$N_{t+1} = N_t + B_t - D_t$$

What distributions should we use for B_t and D_t?

$$B_t \sim \text{Poisson}(N_t \times b)$$

$$D_t \sim \text{Binomial}(N_t, d)$$

Poisson distribution

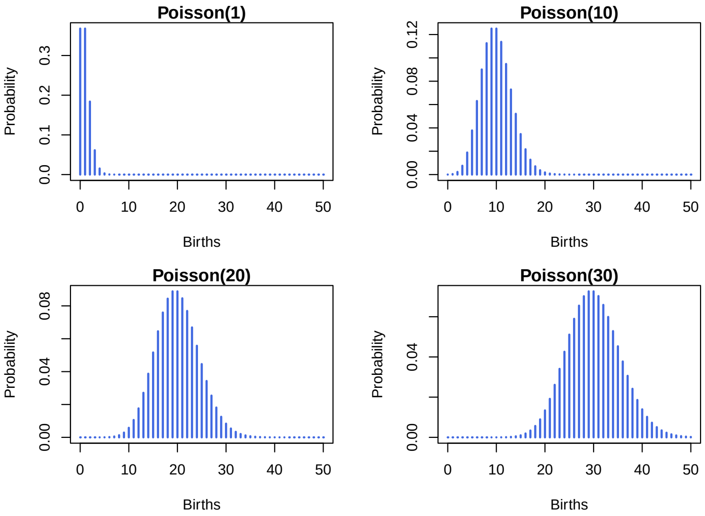
The Poisson distribution is useful for data that are non-negative integers.

It has a single parameter that describes the expected value of the random outcomes.

In stochastic population models, the Poisson distribution can be used to model the number of births (B_t) that occur in a time interval.

$$B_t \sim \text{Poisson}(N_t \times b)$$

Poisson distributions

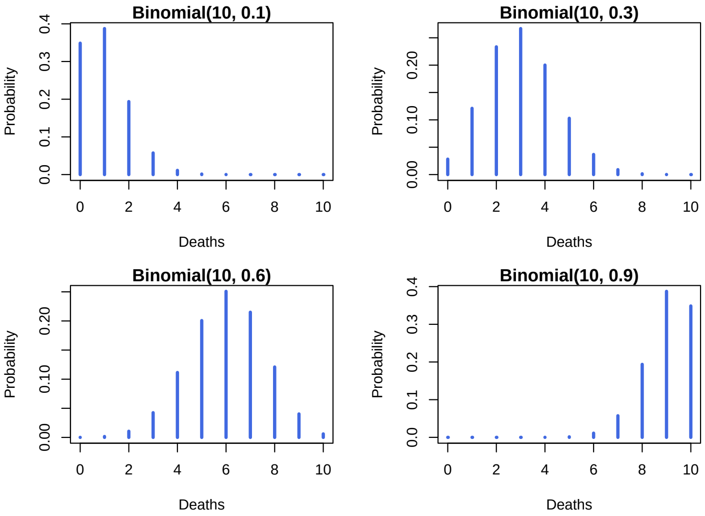


The binomial distribution is also useful for data that are non-negative integers, but it has an upper bound.

In population models, the upper bound is often population size, and we use the model to describe how many individuals die during some time period.

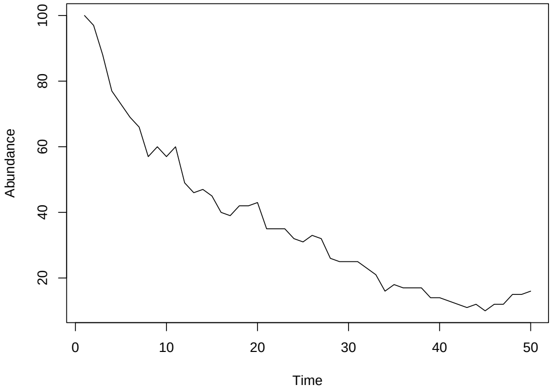
$$D_t \sim \mathrm{Binomial}(N_t, d)$$

Binomial distributions



Poisson-Binomial birth-death model.

$$N_{t+1} = N_t + B_t - D_t$$
$$B_t \sim \mathrm{Poisson}(N_t \times b)$$
$$D_t \sim \mathrm{Binomial}(N_t, d)$$



Purely deterministic models are too rigid.

Purely stochastic models don't describe population processes.

The goal is to develop mechanistic models that represent our biological understanding while allowing for uncertainty.